SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Computer Science and Engineering

ASSESSMENT TOOL 3 – Comparative Analysis

Course Code:	CSA06	**Course Title:**	Design and Analysis of Algorithms
CO Assessed:	**CO4:** Articulation (BL4, BL5)	**Assessment:**	Assessment Tool 3 – Comparative Analysis
Weightage:	10% of CIA	**Total Marks:**	100
CO4 Statement:	Solve optimization problems using dynamic programming and greedy strategies.		
Student Name:	PRITHIYANKA SHREE M	**Reg. No.:**	192521144
Section / Batch:	2025	**Date:**	24-08-2026

Instructions to Students

- Answer the following question. Total Marks: 100.
- Analyze the given questions thoroughly before answering.
- Compare multiple aspects such as methods, results, advantages, and limitations clearly.
- Critically evaluate the strengths, weaknesses, and implications with proper justification.
- Present meaningful insights, interpretations, and well-supported conclusions from the comparisons.

Question Type	Focus Area	Questions
Comparative Analysis	Comparative analysis of Dynamic Programming and Greedy algorithms for optimization problems.	Q1

SECTION A – Comparative Analysis

Given activities with start times [1, 3, 0, 5, 8, 5] and finish times [2, 4, 6, 7, 9, 9], solve using both Greedy and Dynamic Programming approaches. Compare selected activities and efficiency. Evaluate correctness of greedy choice property. Derive insights on scheduling optimization.

Code of Conduct

I PRITHIYANKA SHREE M(192521144) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: PRITHIYANKA SHREE M

RUBRICS FOR CALCULATION

Comparative Analysis					
Criteria	**Weightage**	**Level 4 (Exemplary)**	**Level 3 (Proficient)**	**Level 2 (Developing)**	**Level 1 (Beginning)**
Selection of Studies/Parameters	20%	Selects highly relevant and diverse studies with well-defined comparison parameters	Selects relevant studies with minor gaps in parameter definition	Limited or partially relevant selection	Poor or inappropriate selection of studies
Comparison & Differentiation	25%	Clearly compares multiple aspects (methods, results, limitations) with strong differentiation	Good comparison with minor gaps	Basic comparison with limited depth	No clear comparison; mostly descriptive
Critical Evaluation	25%	Critically evaluates strengths, weaknesses, and implications with strong justification	Provides evaluation with some justification	Limited evaluation; lacks depth	No critical evaluation provided
Synthesis & Insight Generation	20%	Generates meaningful insights and conclusions from comparisons	Some insights derived from comparison	Limited or unclear insights	No insights generated
Clarity	10%	Information is clearly presented (tables/figures) with logical structure	Generally clear with minor issues	Presentation lacks clarity or structure	Poor presentation and difficult to understand

Marks Evaluation:

Criteria	Wt.	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Selection of Studies/Parameters	20%				
Comparison & Differentiation	25%				
Critical Evaluation	25%				
Synthesis & Insight Generation	20%				
Clarity	10%				
Total Marks (100):					

Faculty In-charge	Course Coordinator	Head of Department
Name:	Name:	Name:
Signature: ________________	Signature: ________________	Signature: ________________
Date: ____________________	Date: ____________________	Date: ____________________

Your Code

```
n = int(input())

start = list(map(int, input().split()))
finish = list(map(int, input().split()))

# ---------------- GREEDY ----------------

greedy = []
last_finish = -1

for i in range(n):
    if start[i] >= last_finish:
        greedy.append(i + 1)
        last_finish = finish[i]

print("Greedy:")
for i in range(len(greedy)):
    print(greedy[i], end=" ")

print()

# ---------------- DYNAMIC PROGRAMMING ----------------

dp = []

for i in range(n):
    dp.append(1)

for i in range(n):
    for j in range(i):
        if finish[j] <= start[i]:
            if dp[j] + 1 > dp[i]:
                dp[i] = dp[j] + 1

maximum = dp[0]
last = 0

for i in range(1, n):
    if dp[i] > maximum:
        maximum = dp[i]
        last = i

print("DP:")
print("Maximum activities:", maximum)
```

INPUT AND OUTPUT:

Input

```
6
1 3 0 5 8 5
2 4 6 7 9 9
```

Output

```
Greedy:
1 2 4 5
DP:
Maximum activities: 4
```